

A

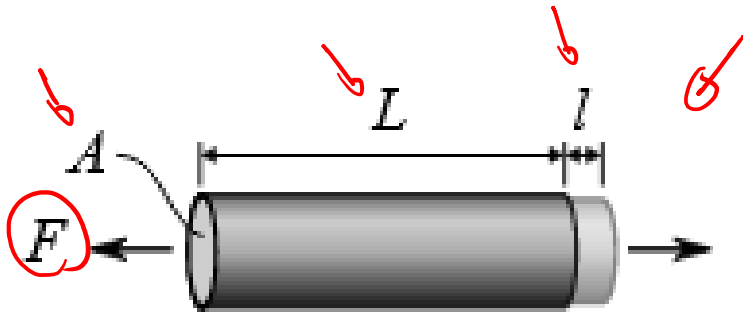
STM - Exercise 1

Stefano De La Pierre

Exercise 1

Use figure 1 to:

- a) define which test and which class of materials is referred to and why;
- b) calculate elastic modulus,
- c) Calculate the yield strength (0.2 % offset) and the tensile strength,
- d) Calculate the elongation at fracture for a 50 mm long specimen (write on curves data you use).
- e) Calculate the length of the specimen when you keep it loaded in A.



$$\sigma = \frac{F}{A}$$

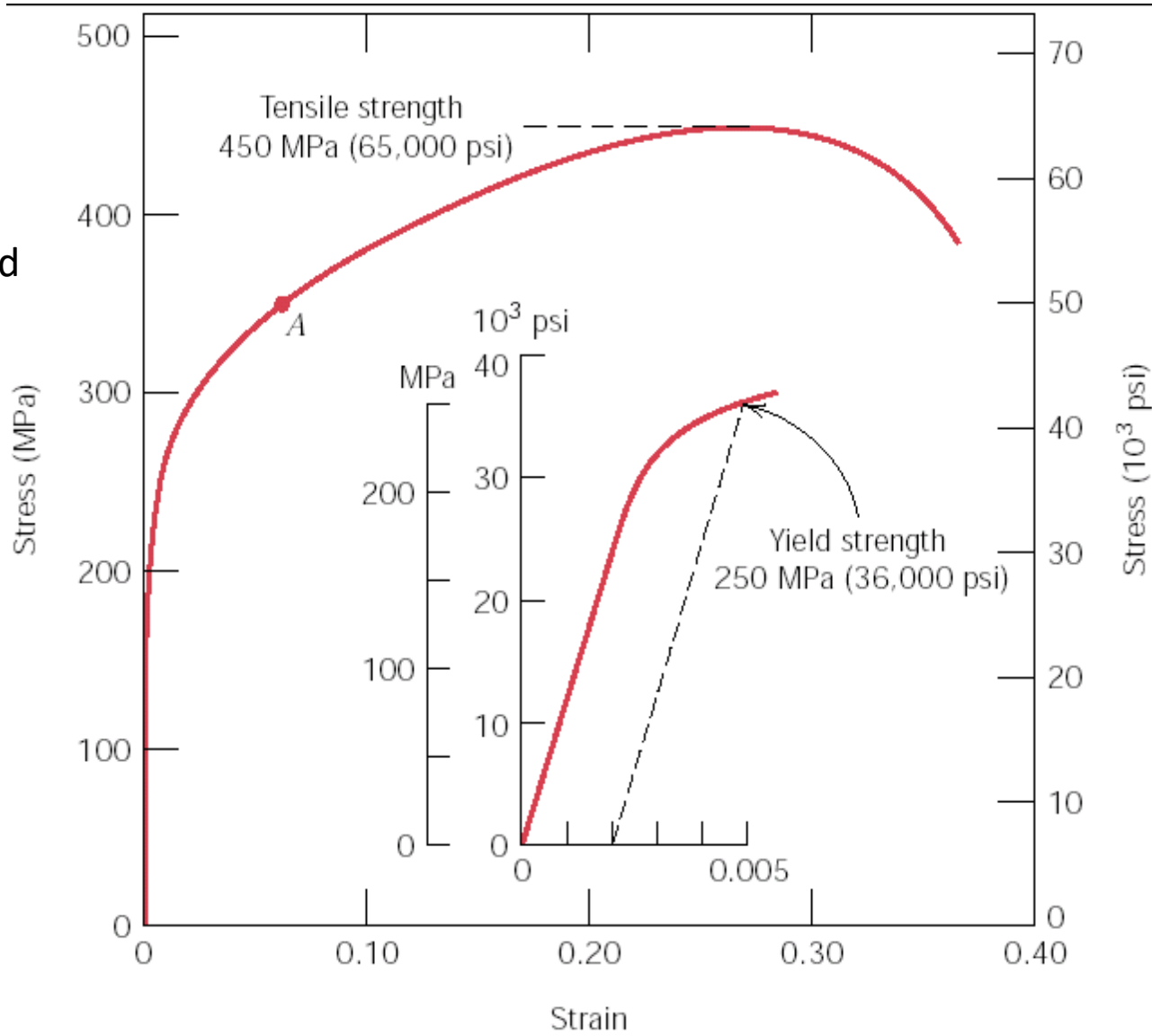
where

- σ stress [Pa]
- F applied force [N]
- A cross-sectional area [m²]

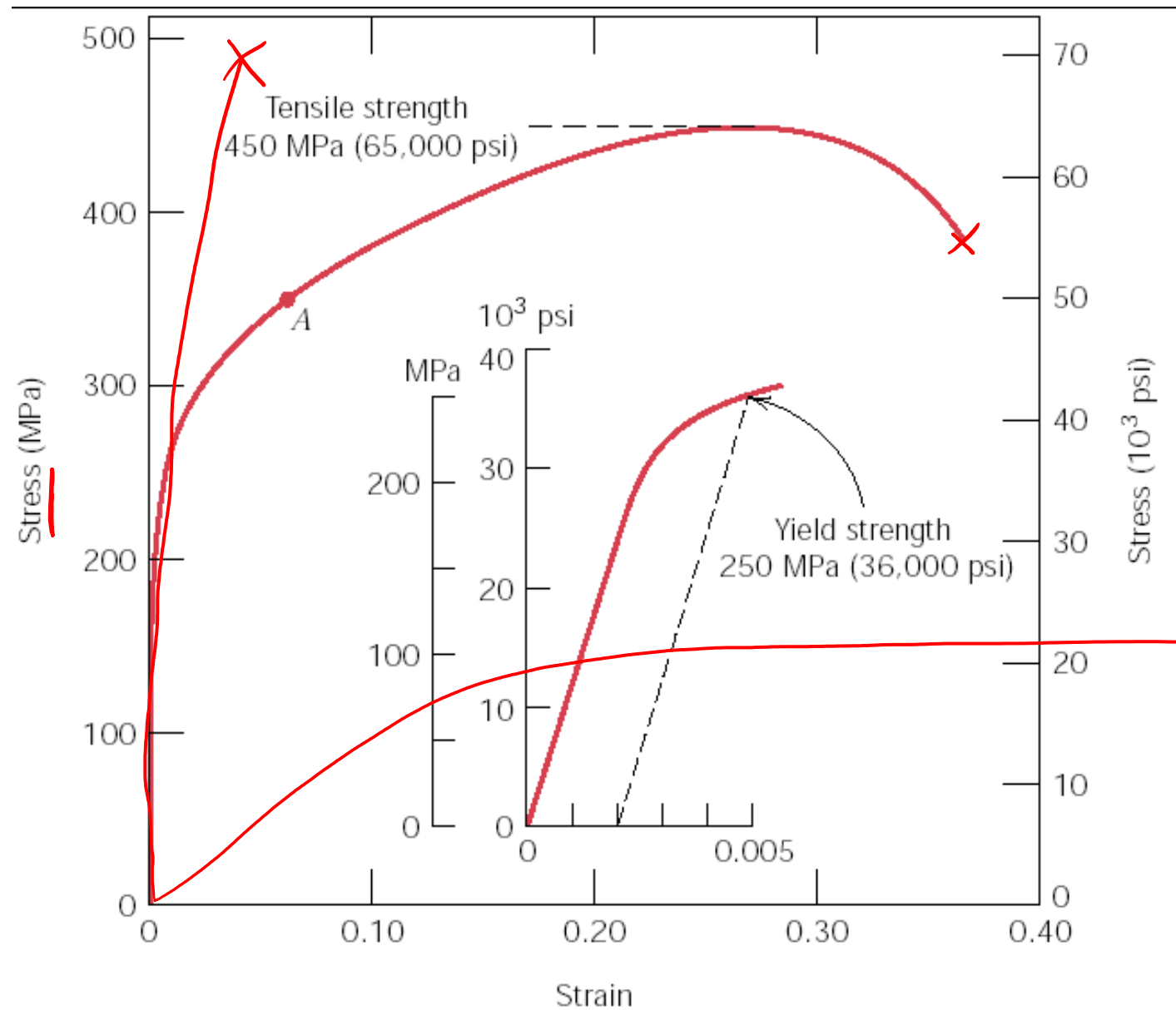
$$\epsilon = \frac{\Delta L}{L_0}$$

where

- ϵ strain
- ΔL total elongation [m]
- L_0 original length [m]

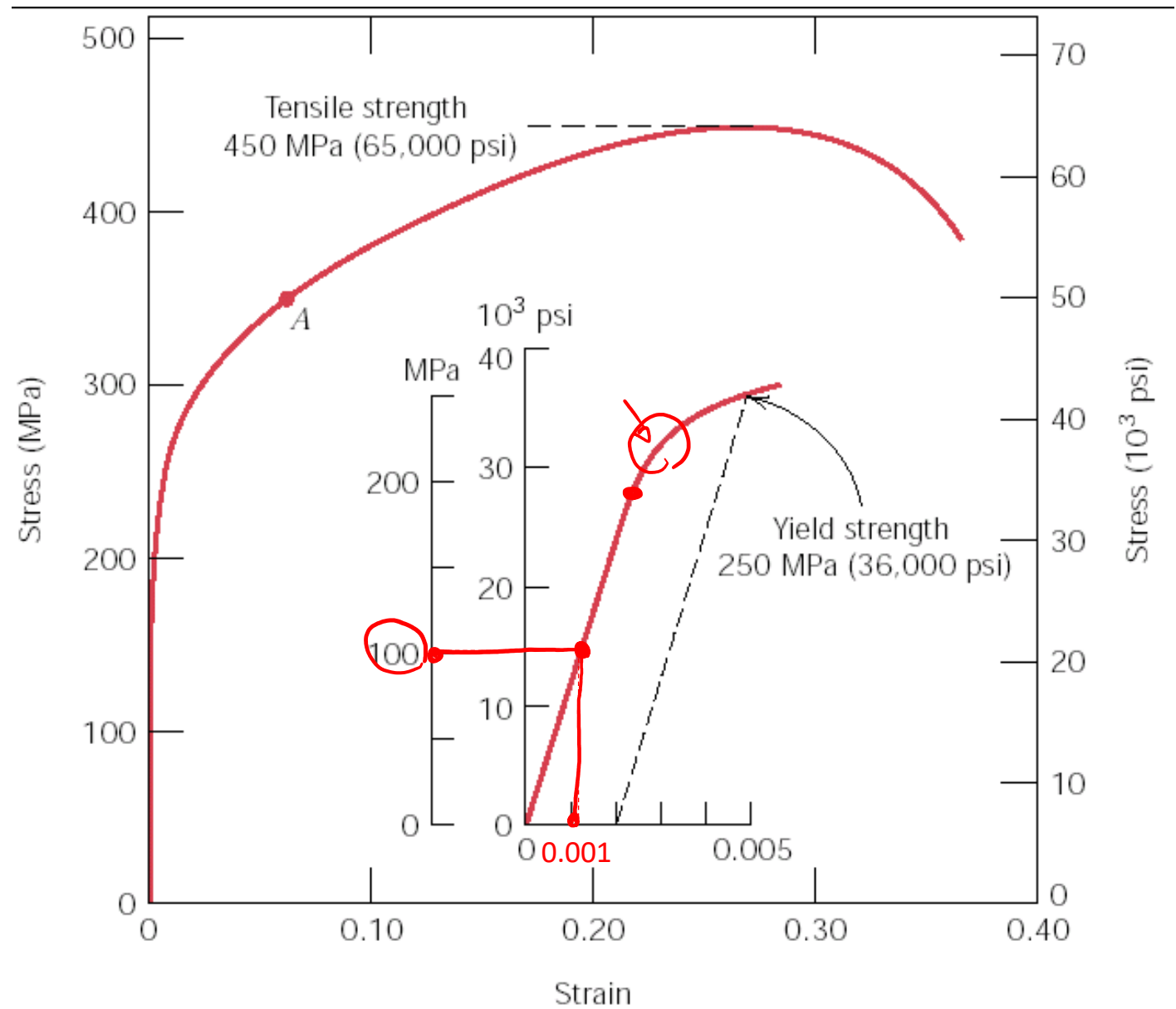


Exercise 1a: Use figure 1 to define which test and which class of materials is referred to and why



Typical stress-strain curve from tensile test
Ductile behavior and stress-strain values typical of
metallic materials

Exercise 1b: Calculate elastic modulus



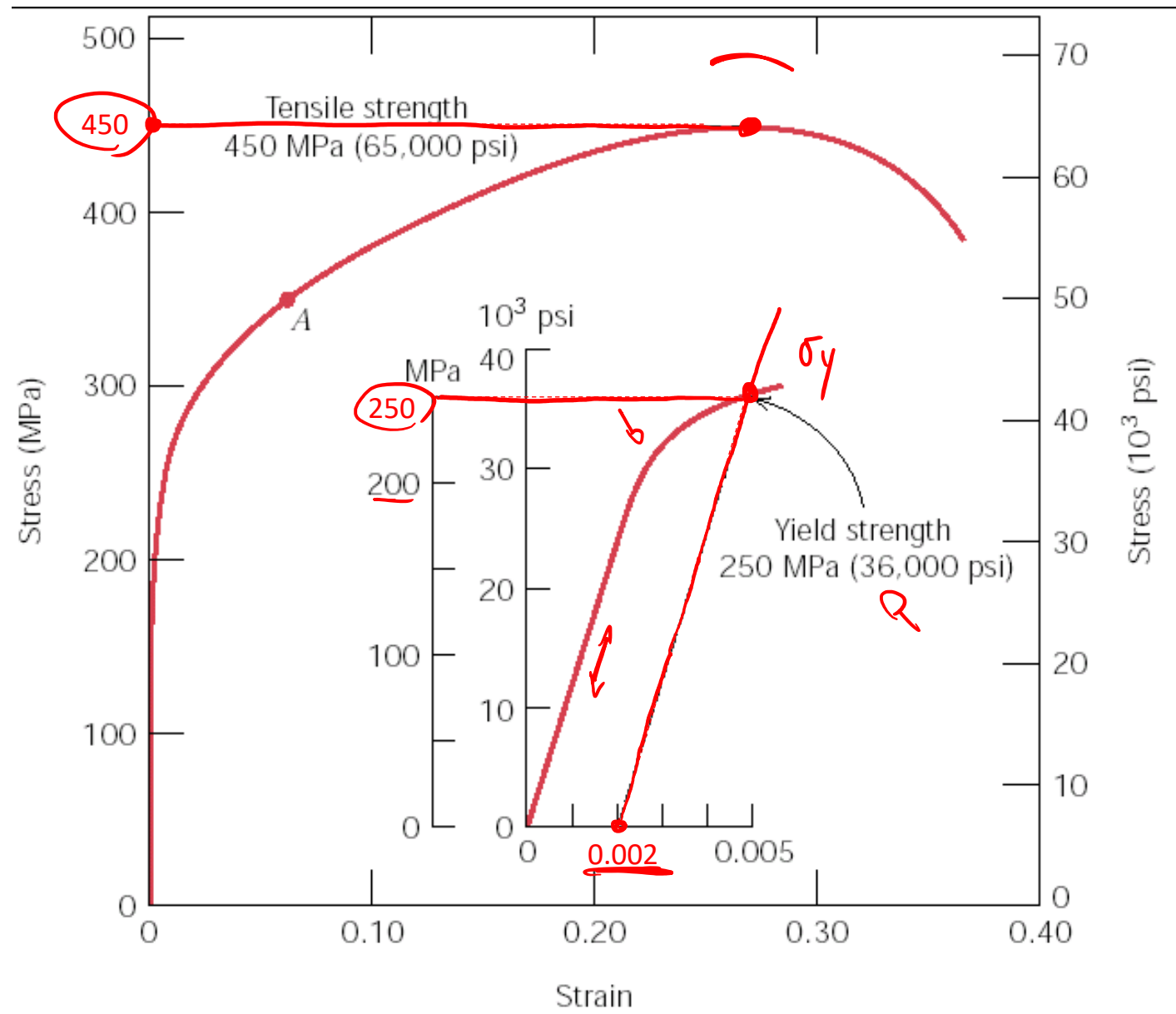
$$E = \frac{\sigma}{\epsilon}$$

$\sigma = 100 \text{ MPa}$

$\epsilon = 0,001$

$$E = \frac{100 \text{ MPa}}{0,001} = 100.000 \text{ MPa} = 100 \text{ GPa}$$

Exercise 1c: calculate yield strength (0.2 % offset) and tensile strength



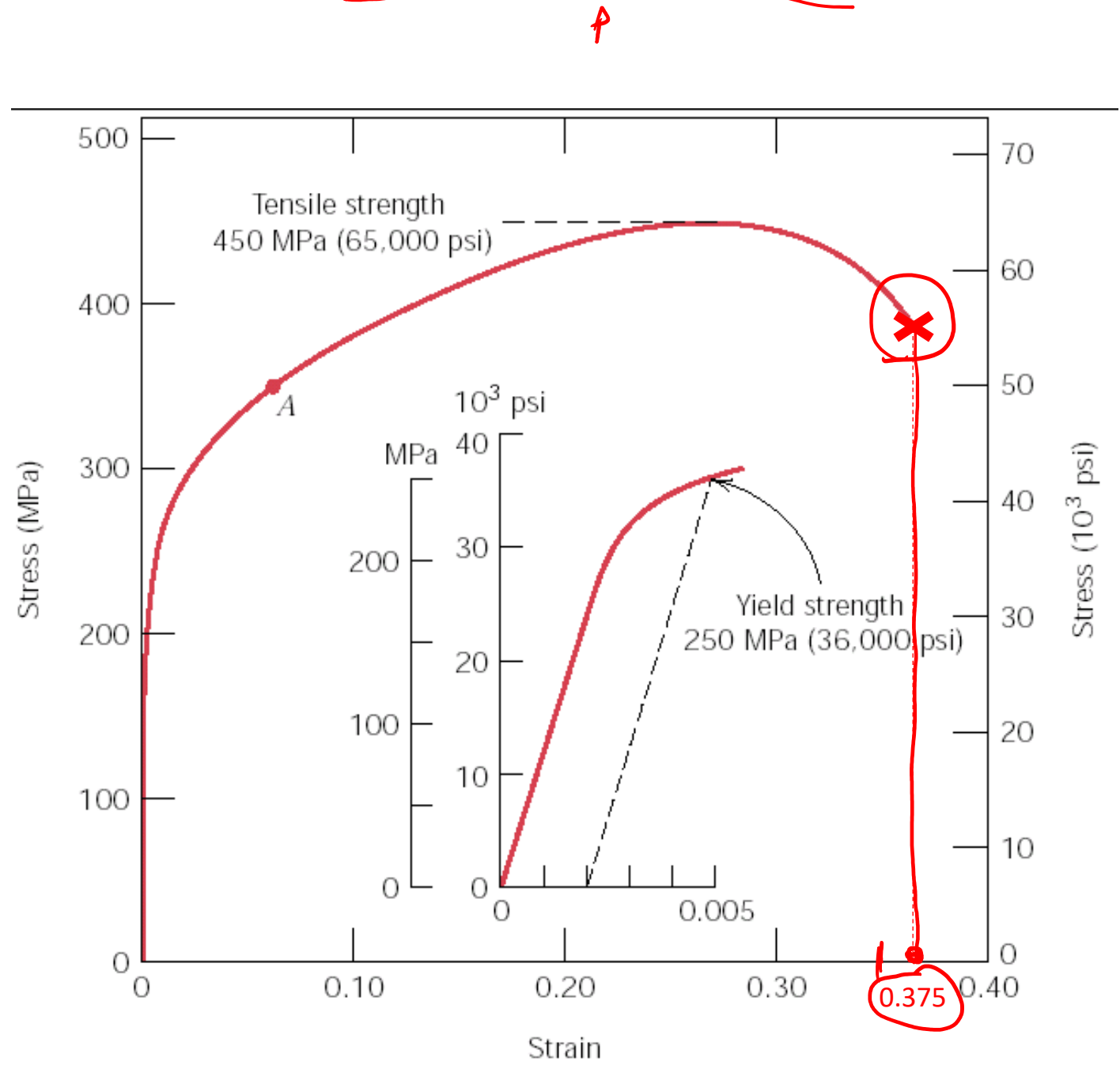
$$\sigma_y = 250 \text{ MPa}$$

$$\epsilon = 0.2\% = 0.002$$

$$\sigma_{\epsilon} = \sigma_{\max} = \text{UTS}$$

$$\sigma_{\max} = 450 \text{ MPa}$$

Exercise 1d: Calculate elongation at fracture for a 50 mm long specimen (write on curves data you use).



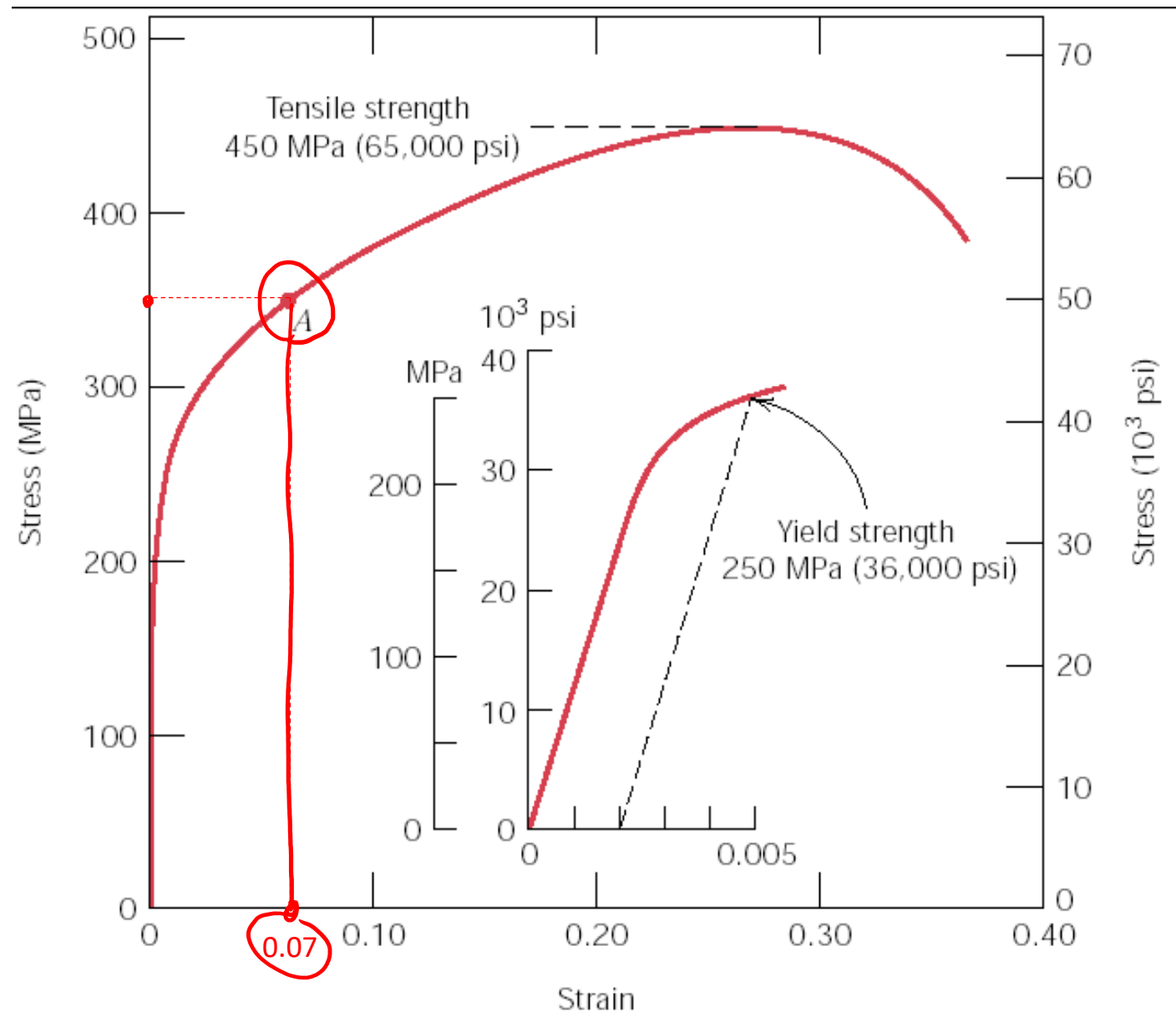
$$\epsilon = \frac{\Delta l}{L_0}$$

$$L_0 = 50 \text{ mm}$$

$$\epsilon = 0,375$$

$$\Delta l = \epsilon \cdot L_0 = 0,375 \cdot 50 \text{ mm} = 18,75 \text{ mm}$$

Exercise 1e- Calculate the length of the specimen when you keep it loaded in A.



$\epsilon = \frac{\Delta l}{L_0}$ $L_0 = 50 \text{ mm}$
 $\epsilon = 0,07$

$\Delta l = \epsilon \cdot L_0 = 0,07 \cdot 50 \text{ mm} = 3,5 \text{ mm}$

$L_A = L_0 + \Delta l = 50 \text{ mm} + 3,5 \text{ mm} =$

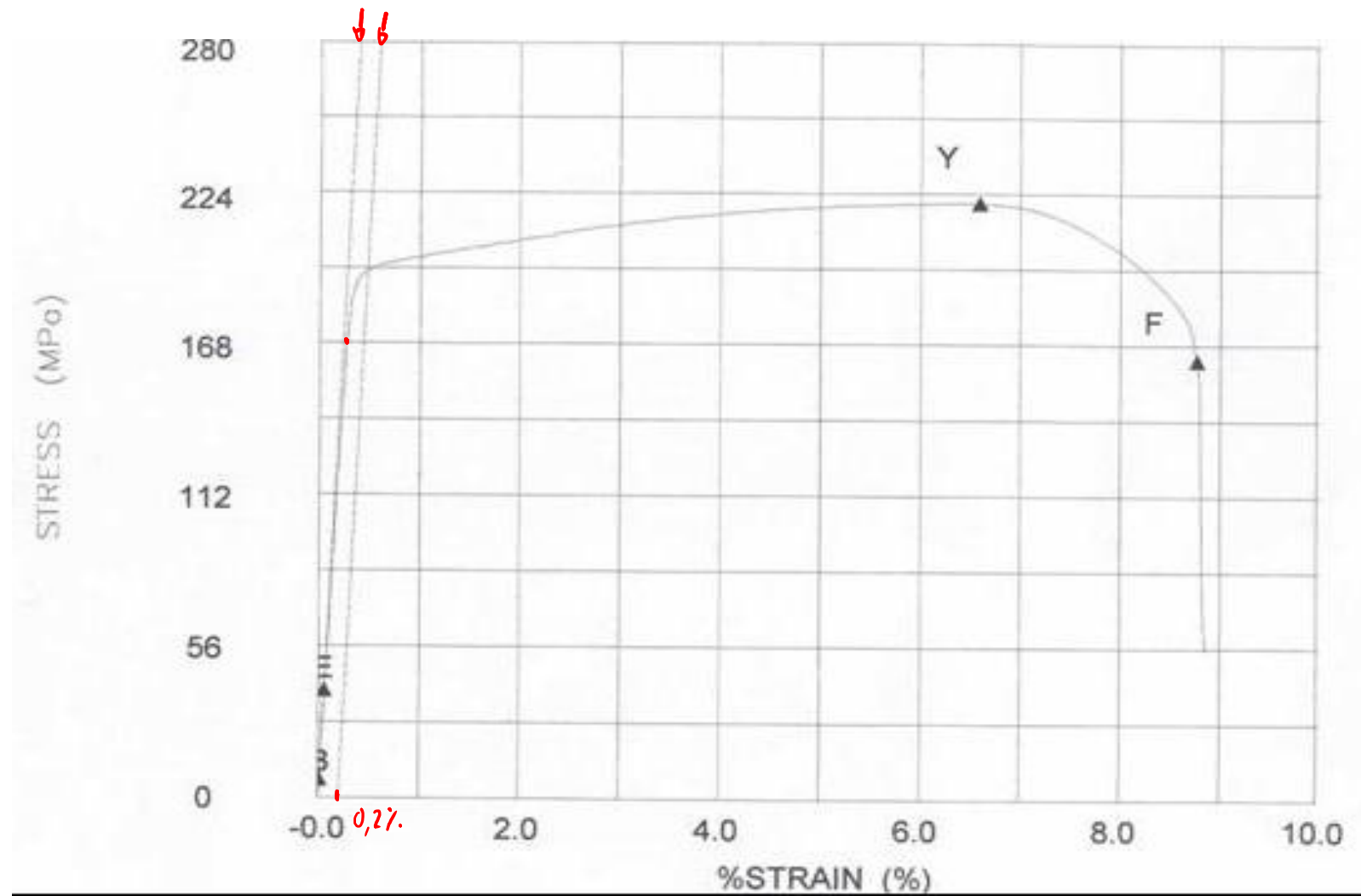
$L_A = 53,5 \text{ mm}$

If you keep the sample loaded no recovery of elastic deformation occurs

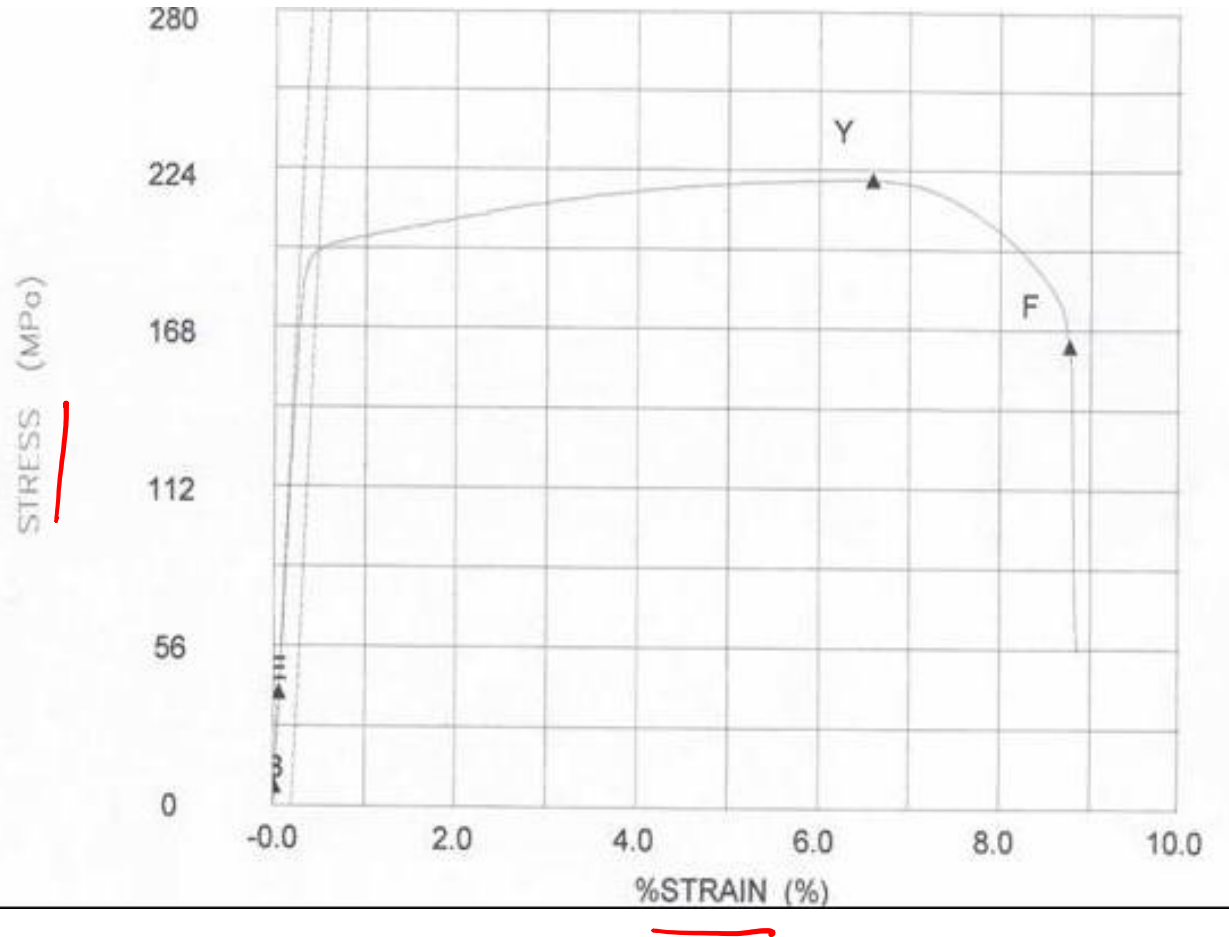
Exercise 2: Use the graph below to define

- which test and which class of materials is referred to and why;
- to calculate elastic modulus,
- To calculate the yield strength (0,2 % strain) and the tensile strength,
- To calculate the elongation at fracture for a 50 mm long specimen;
- To calculate the elongation of this sample when loaded at 200 MPa, then unloaded.

(write on graph stress and strain
you use to calculate).



Exercise 2a: Define which test and which class of materials is referred to and why



TENSILE

METALLIC

Typical stress-strain curve from tensile test
Ductile behavior and stress-strain values typical
of metallic materials

Exercise 2b: calculate elastic modulus

140 MPa 0,1%

1

$$E = \frac{\sigma}{\epsilon}$$

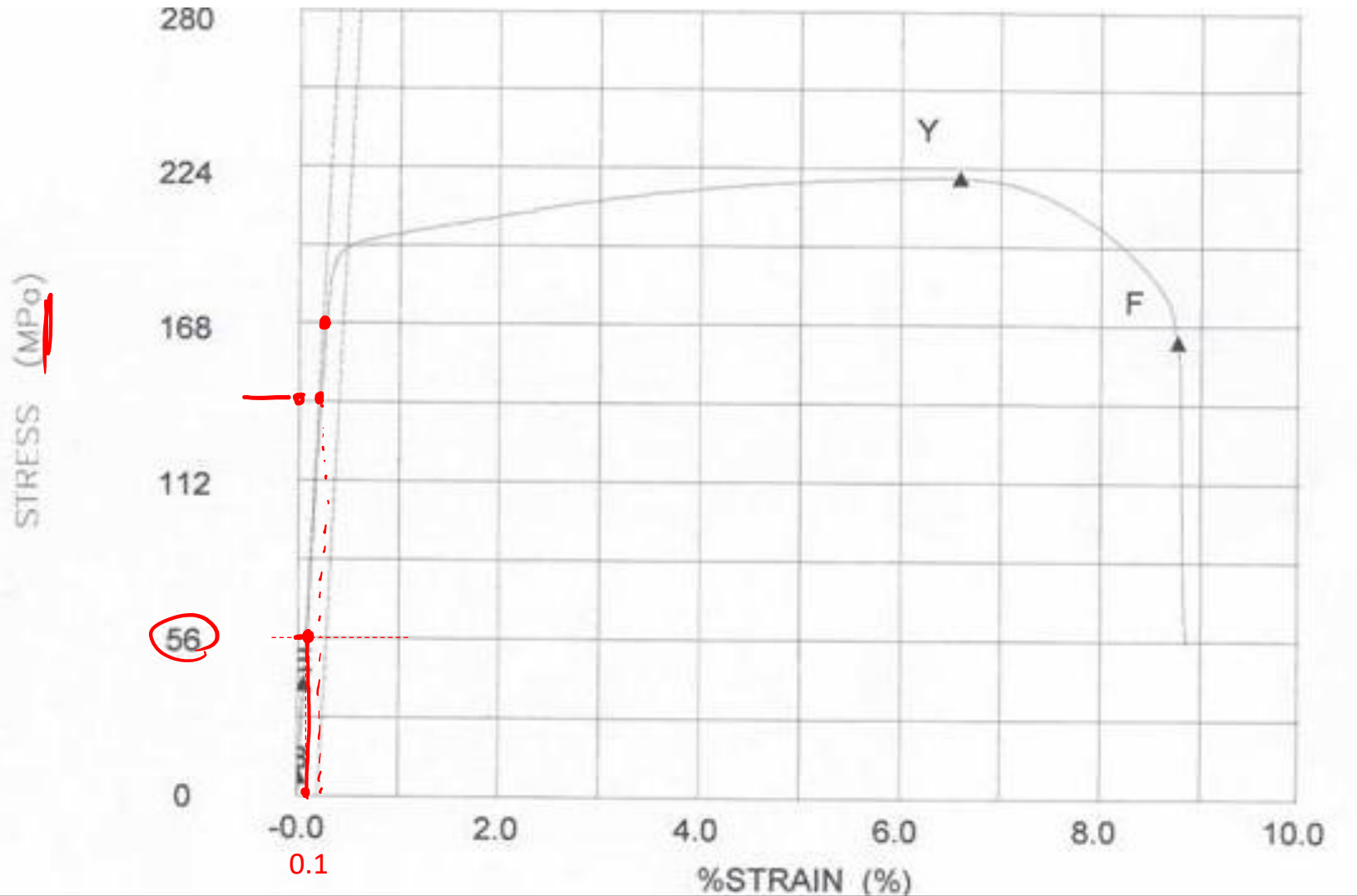
..

$$\sigma = 56 \text{ MPa}$$

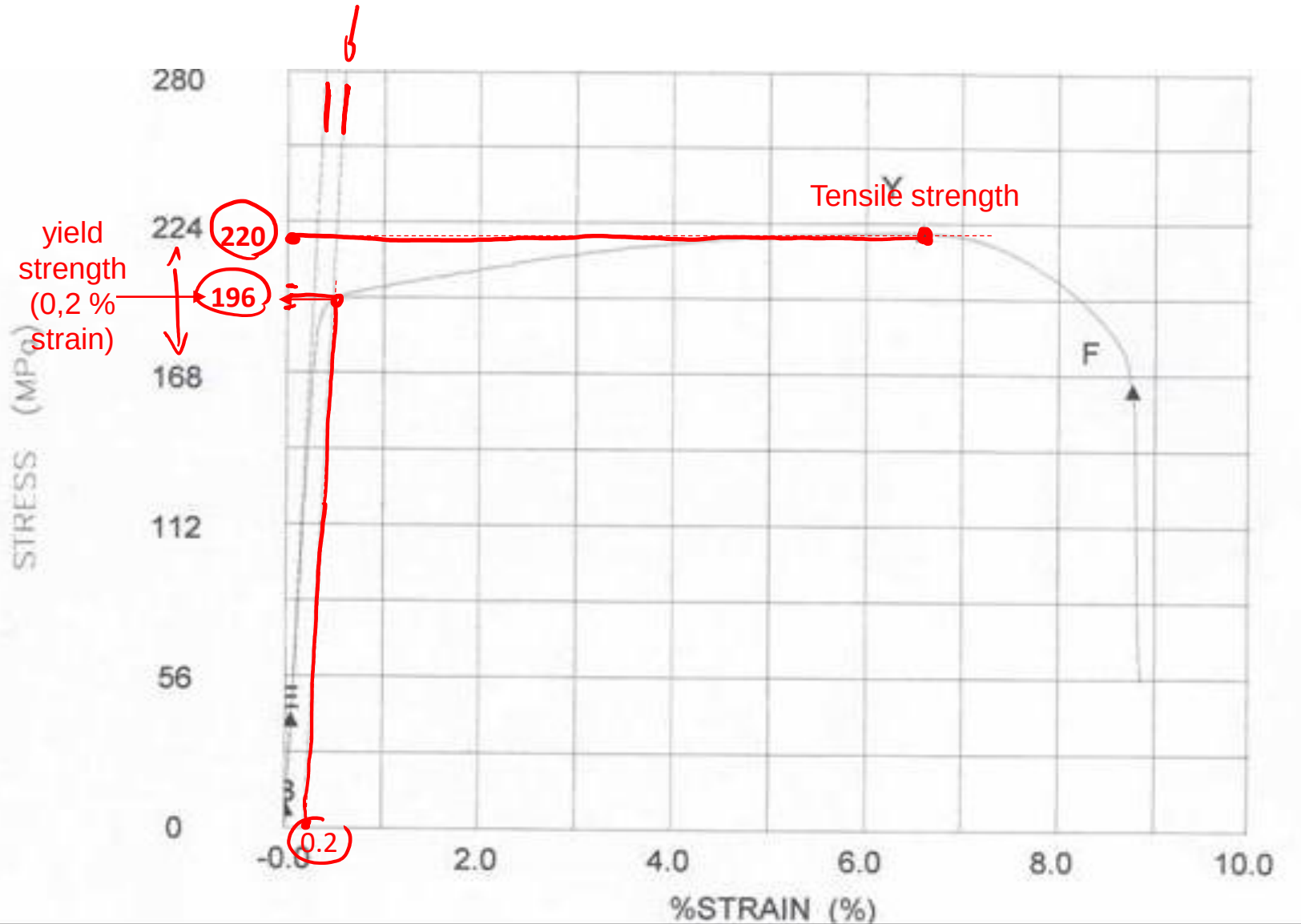
$$\epsilon = 0,1\% = 0,001$$

$$E = \frac{56 \text{ MPa}}{0,001} = 56.000 \text{ MPa} = 56 \text{ GPa}$$

$$E = 70 \text{ GPa}$$



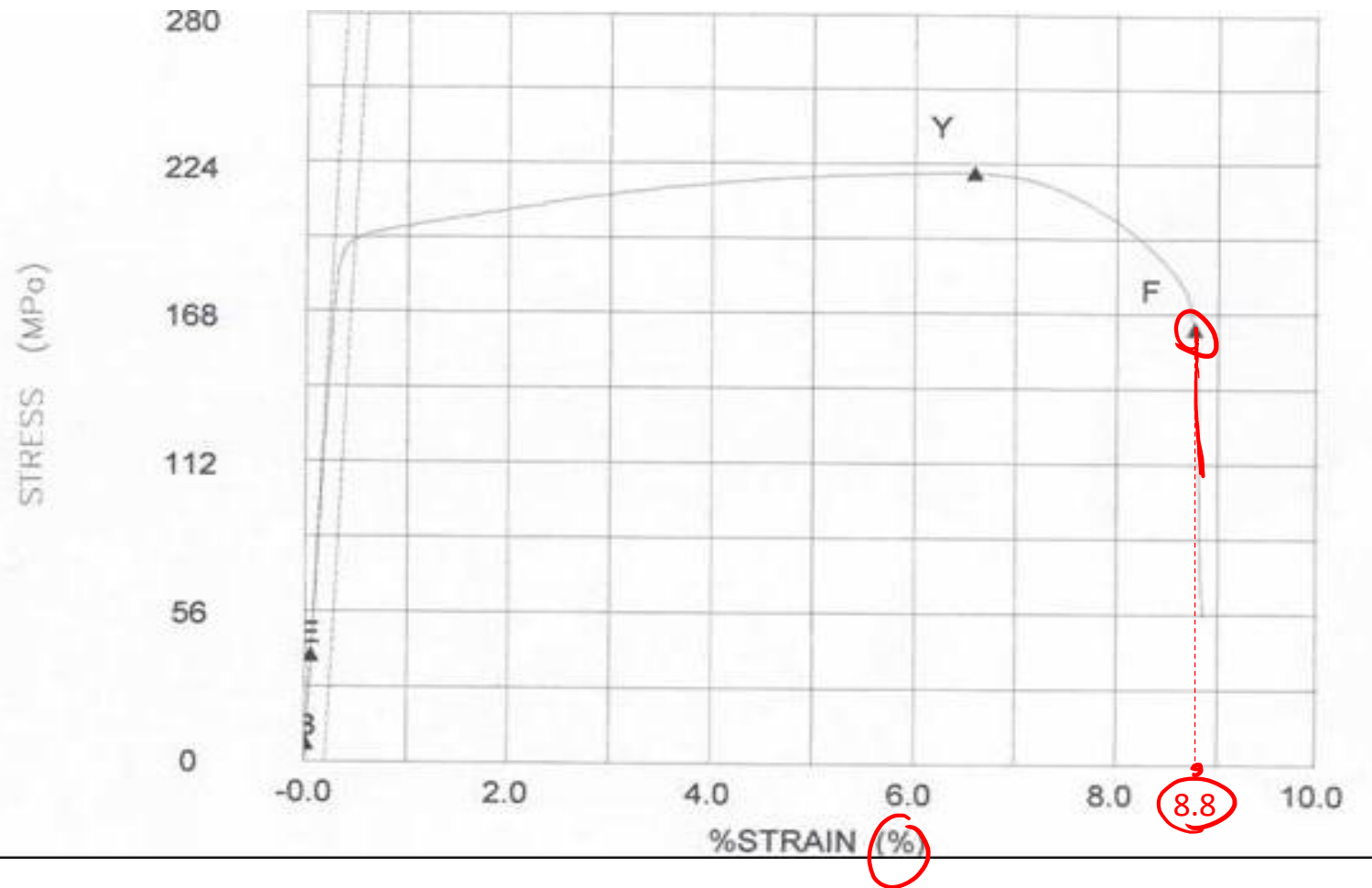
Exercise 2c: calculate yield strength (0,2 % strain) and tensile strength



$$\sigma_y = 196 \text{ MPa}$$

$$\sigma_t = 220 \text{ MPa}$$

Exercise 2d: Calculate elongation at fracture for a 50 mm long specimen

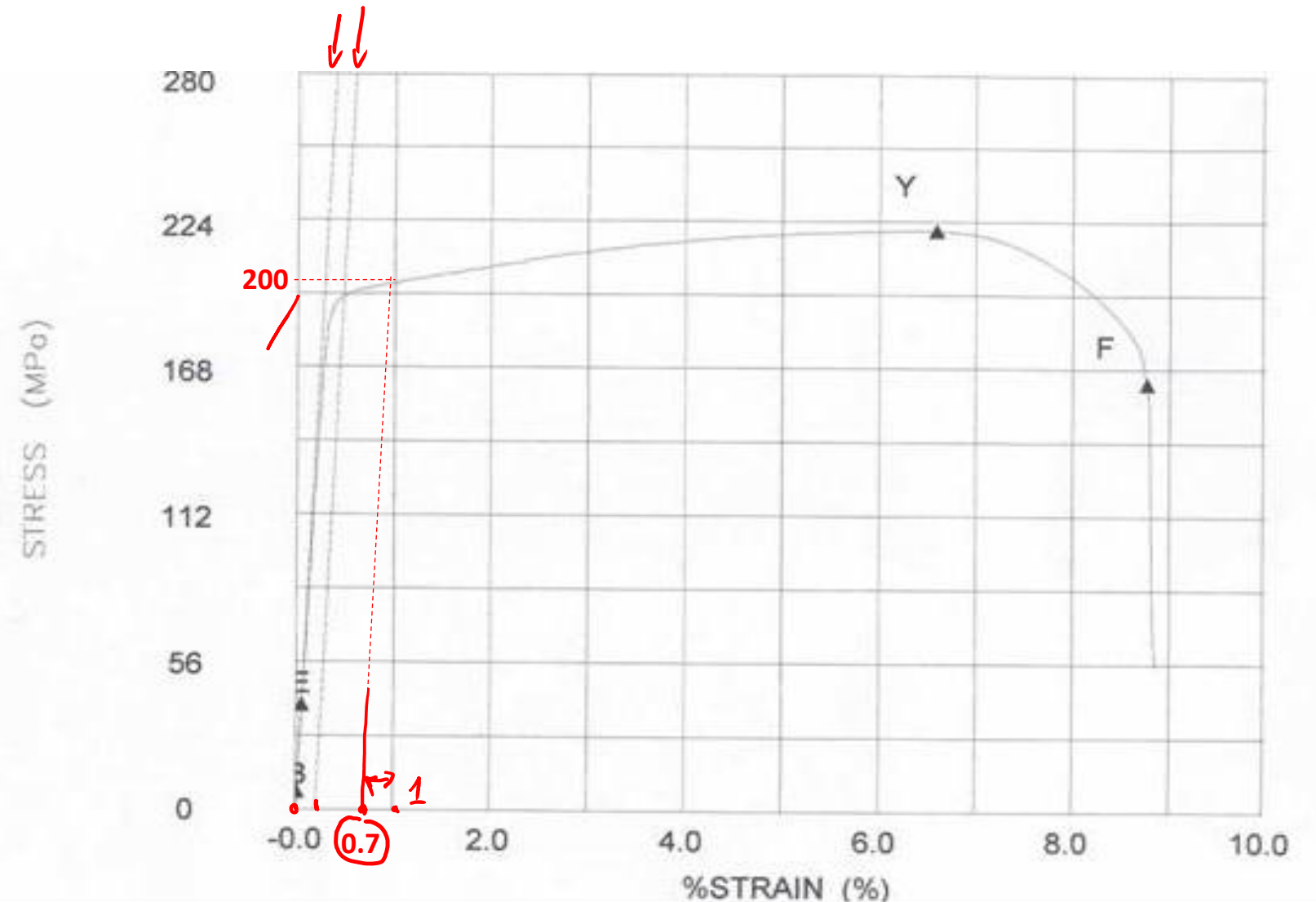


4,4 mm
 $\Delta l =$

$$\Delta l = \epsilon \cdot L_0 = 0,088 \cdot 50 \text{ mm} = 4,4 \text{ mm}$$

$$\epsilon = 8,8\% = 0,088$$

Exercise 2e: Calculate the elongation of this sample when loaded at 200 MPa, then unloaded. (write on graph stress and strain you use to calculate).



$$\Delta l = \epsilon \cdot L_0 = 0,007 \cdot 50 \text{ mm} = 0,35 \text{ mm}$$

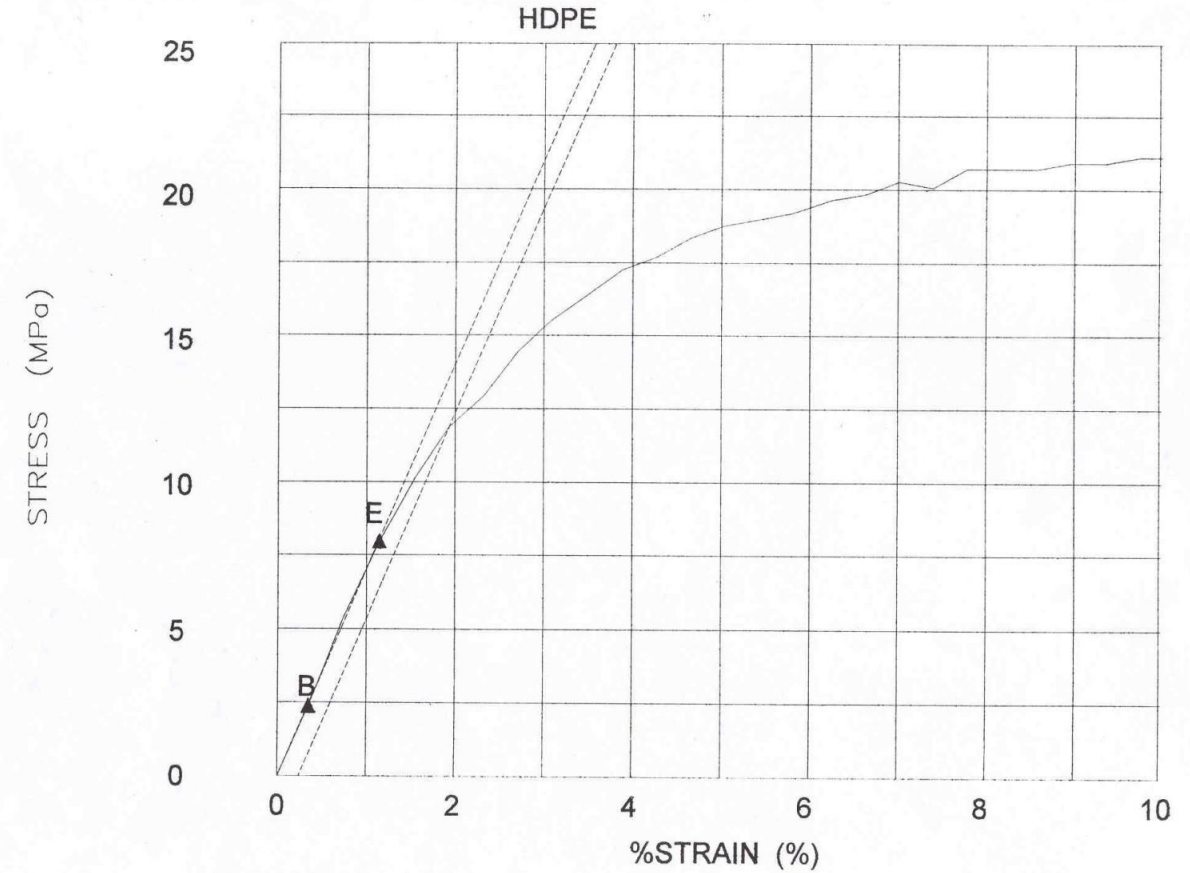
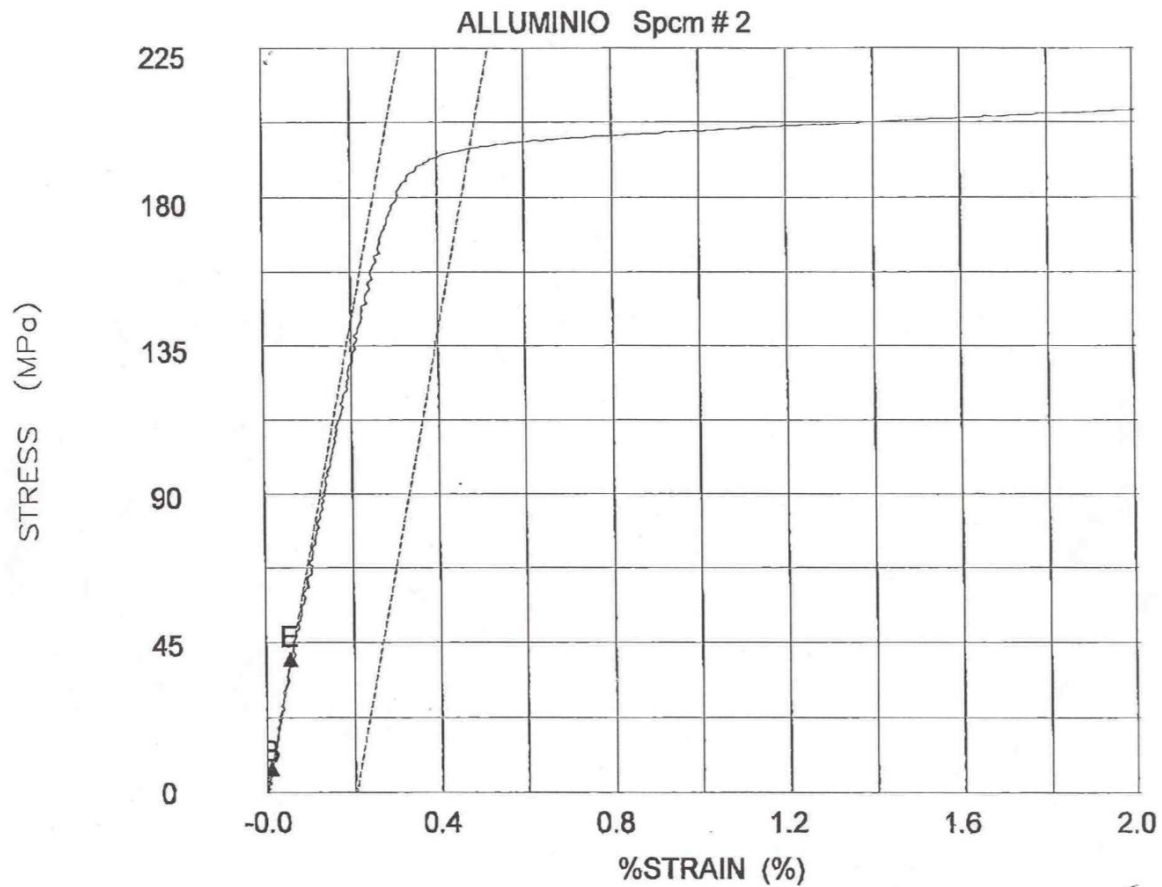
$$\epsilon = 0,7\% = 0,007$$

Loading and unloading →

elastic deformation recovered,
only permanent deformation
present

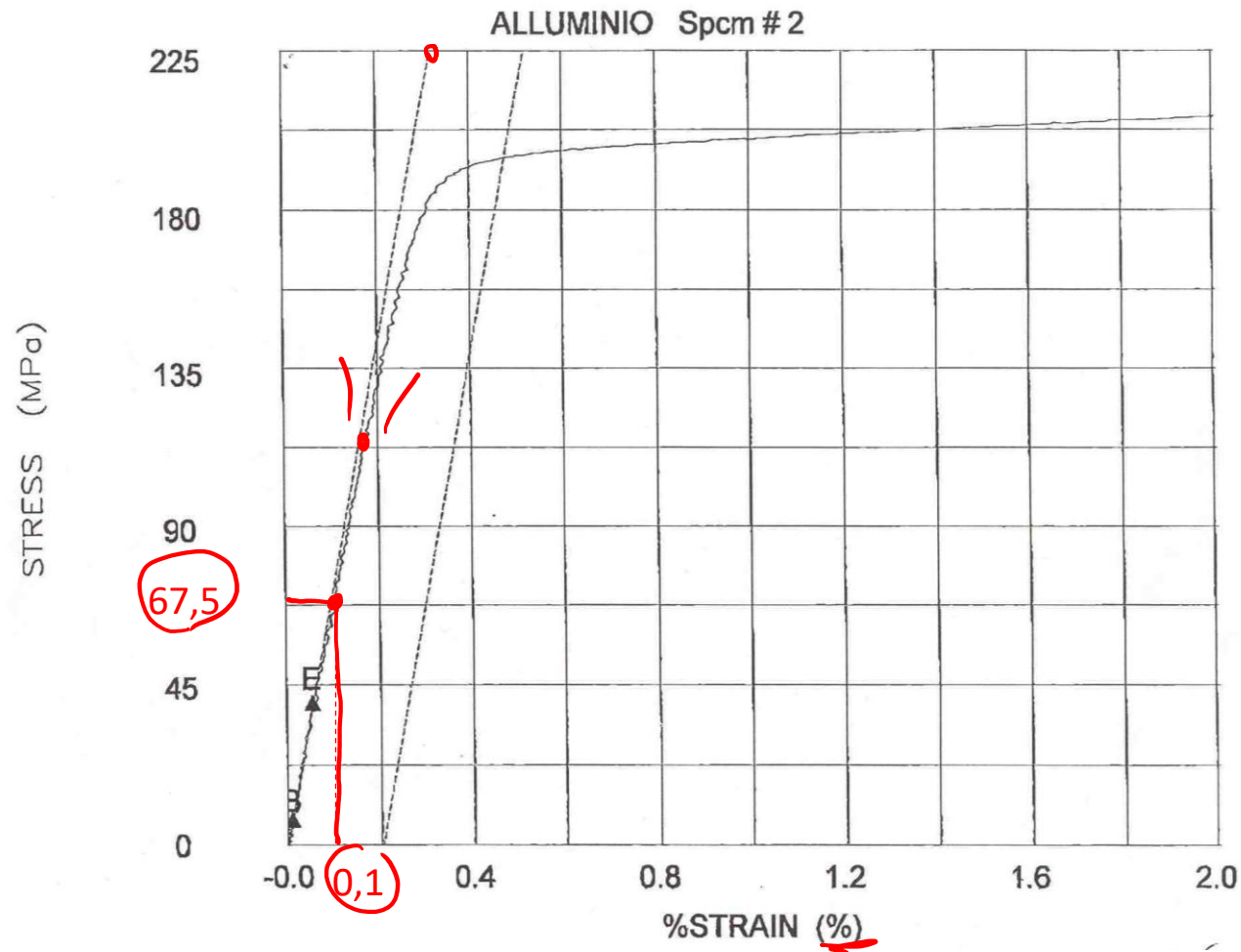
Exercise 3: Determine the Elastic Modulus of Aluminum and High Density Polyethylene (HDPE) from the graphs below. Comment the obtained results.

$$E = \frac{\sigma}{\epsilon}$$



Exercise 3 – case 1: Aluminum

Elastic modulus (E) = slope of the linear (elastic) tract of the σ - ϵ curve

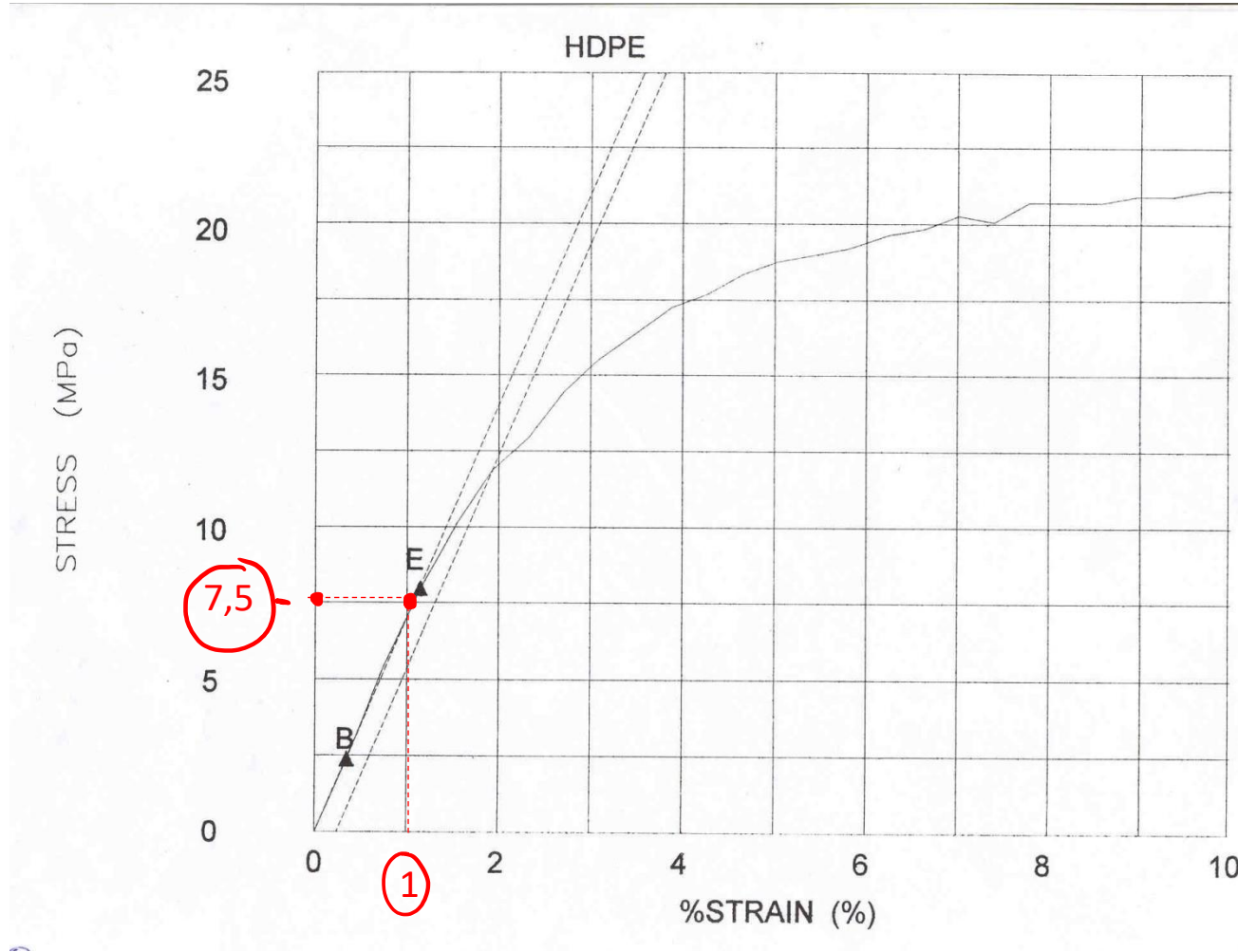


$$\epsilon\% = 0,1\% \rightarrow \epsilon = 0,001$$

$$E = \sigma / \epsilon = 67,5 \text{ MPa} / 0,001 = 67500 \text{ MPa} \rightarrow 67,5 \text{ GPa}$$

Exercise 3 – case 2: HDPE

Elastic modulus (E) = slope of the linear (elastic) tract of the σ - ϵ curve



$$\epsilon\% = 1\% \rightarrow \epsilon = 0,01$$

$$E = \sigma / \epsilon = 7,5 \text{ MPa} / 0,01 =$$

750 MPa → **0,75 GPa**

Exercise 3 comparison Aluminum – HDPE

$E(\text{Al}) = 67.5 \text{ GPa}$

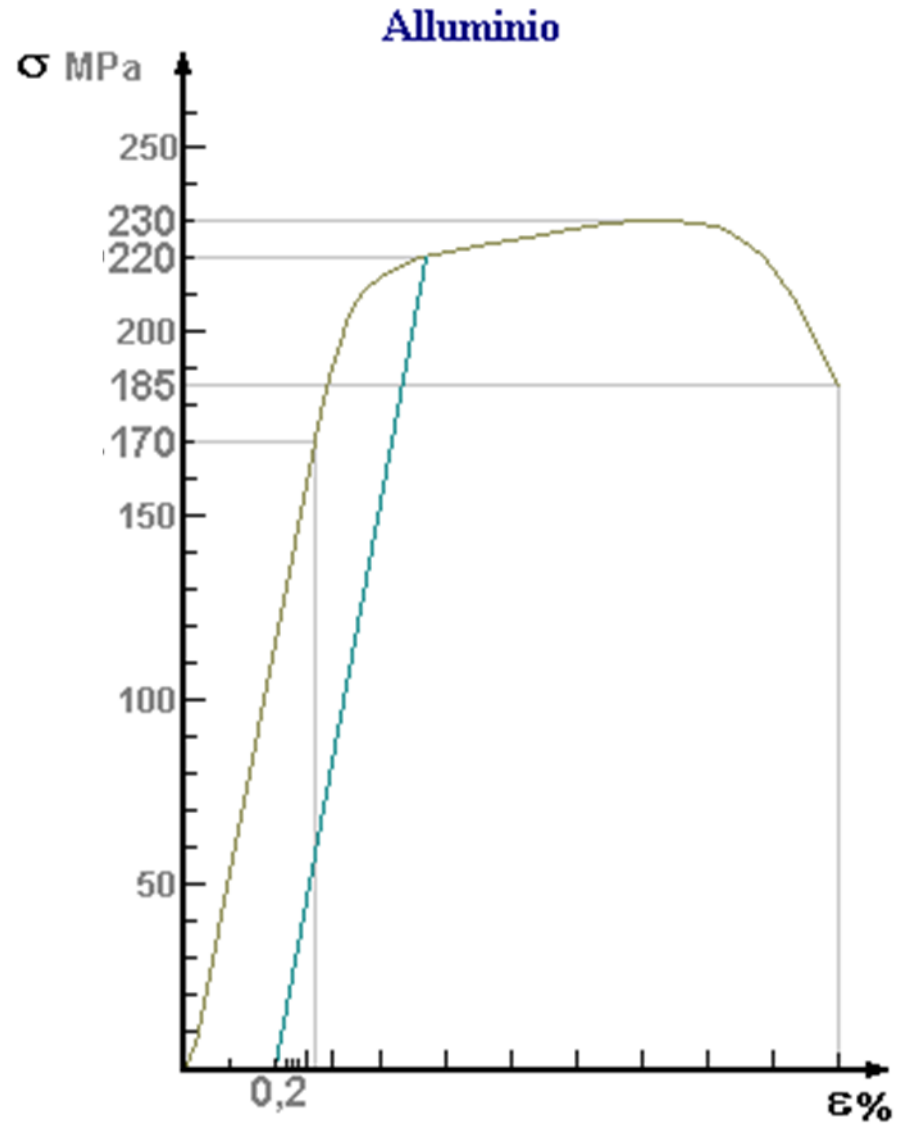
$E(\text{HDPE}) = 0.75 \text{ GPa}$



1 order of magnitude difference

Elastic modulus depends on the bonding strength: metallic bond vs van der Waals interactions

Quiz 1

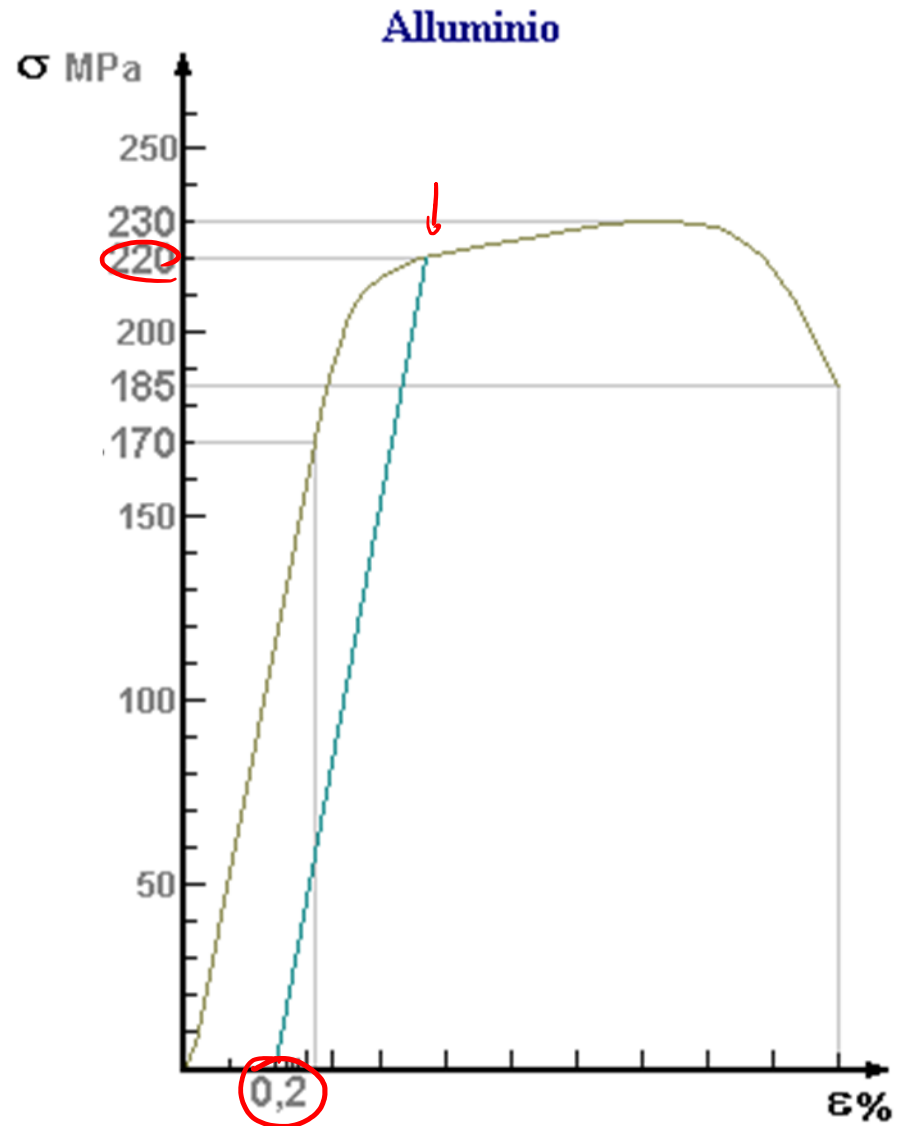


Consider the σ - ϵ curve reported in the figure.

The yield strength (0.2 % offset) is:

- a) 230 MPa
- b) 220 MPa
- c) 185 MPa

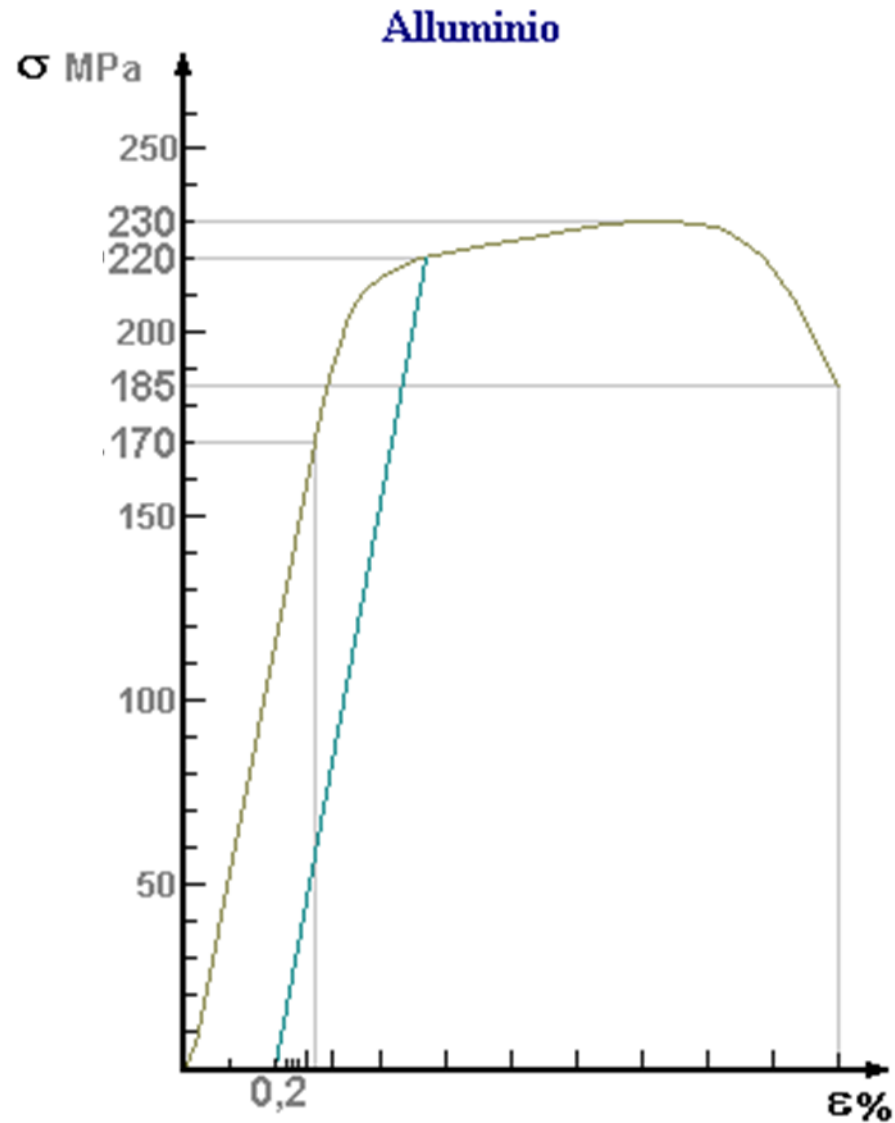
Quiz 1



Consider the σ - ϵ curve reported in the figure. The yield strength (0.2 % offset) is:

- a) 230 Mpa
- b) 220 Mpa**
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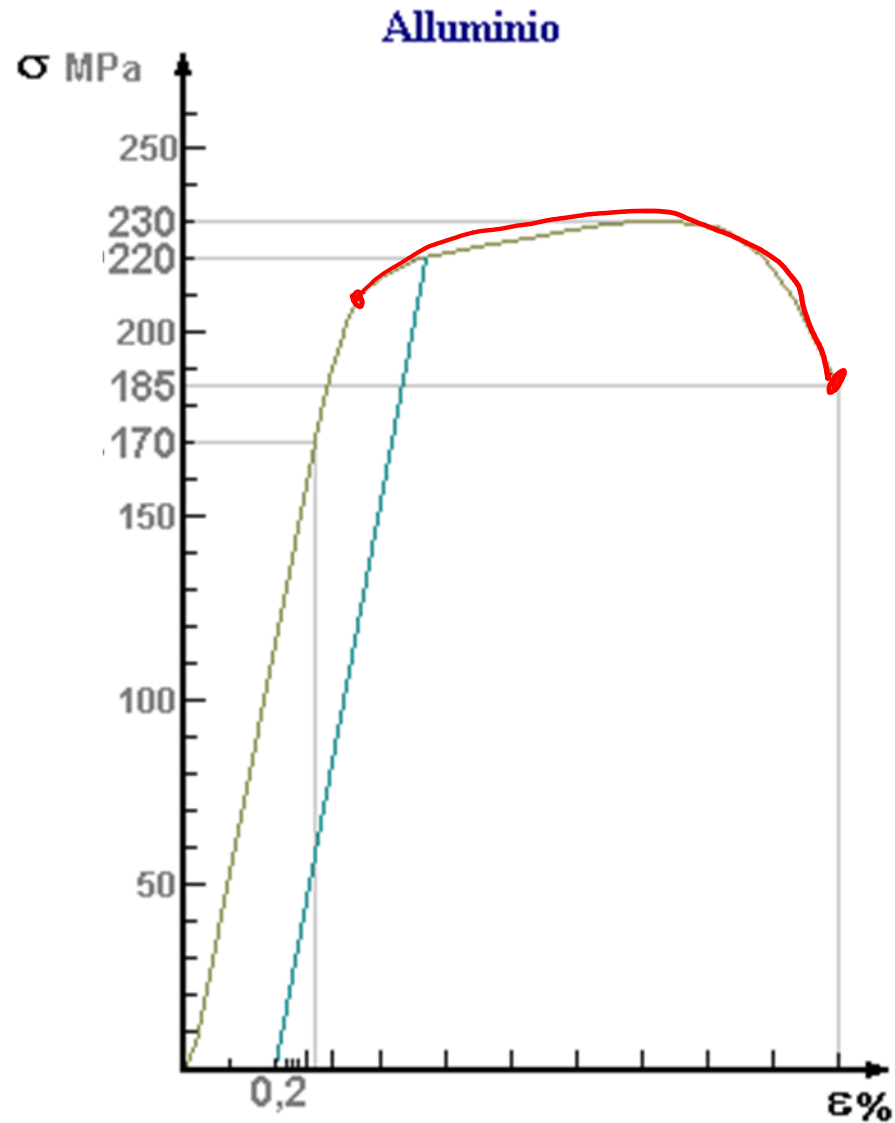
Quiz 2



The σ - ϵ curve in figure comes from the tensile test of an aluminum specimen. Select the right sentence (only one):

- a) The specimen underwent plastic deformation
- b) The specimen underwent fragile fracture
- c) The specimen did not have permanent deformations

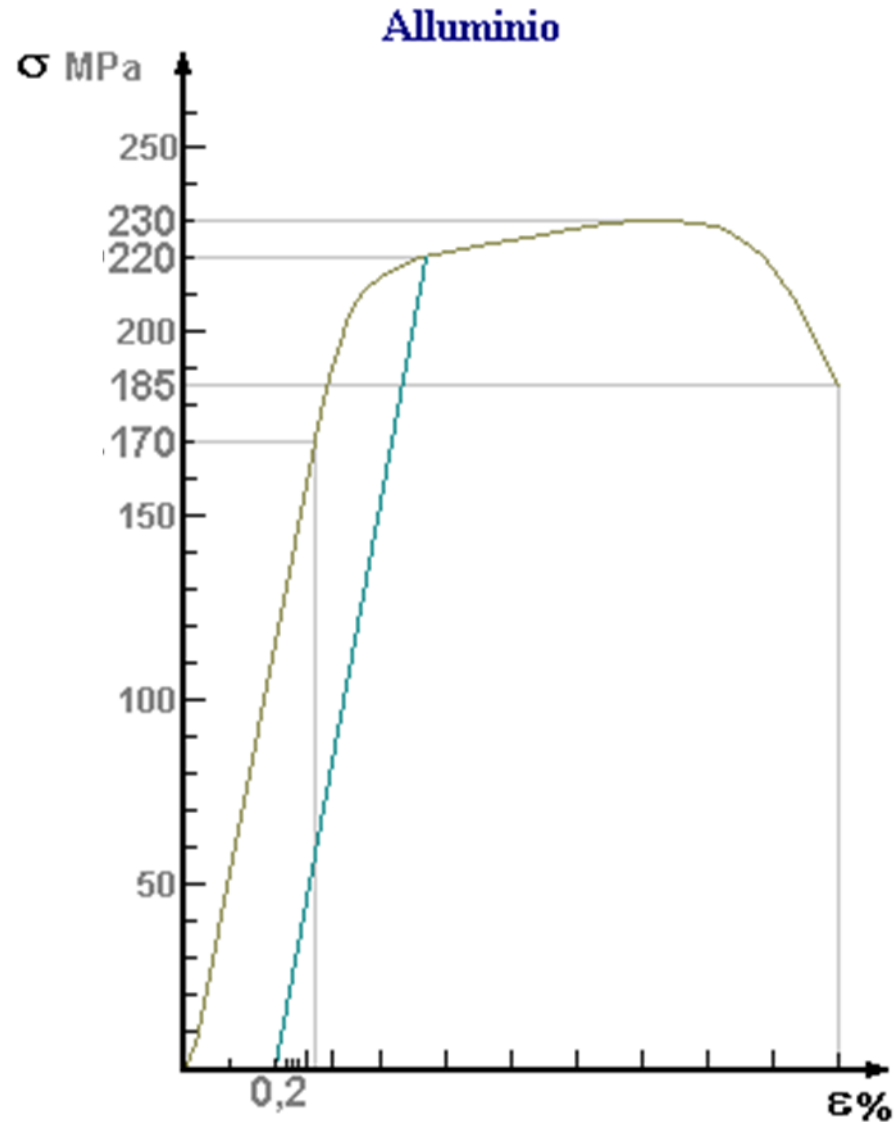
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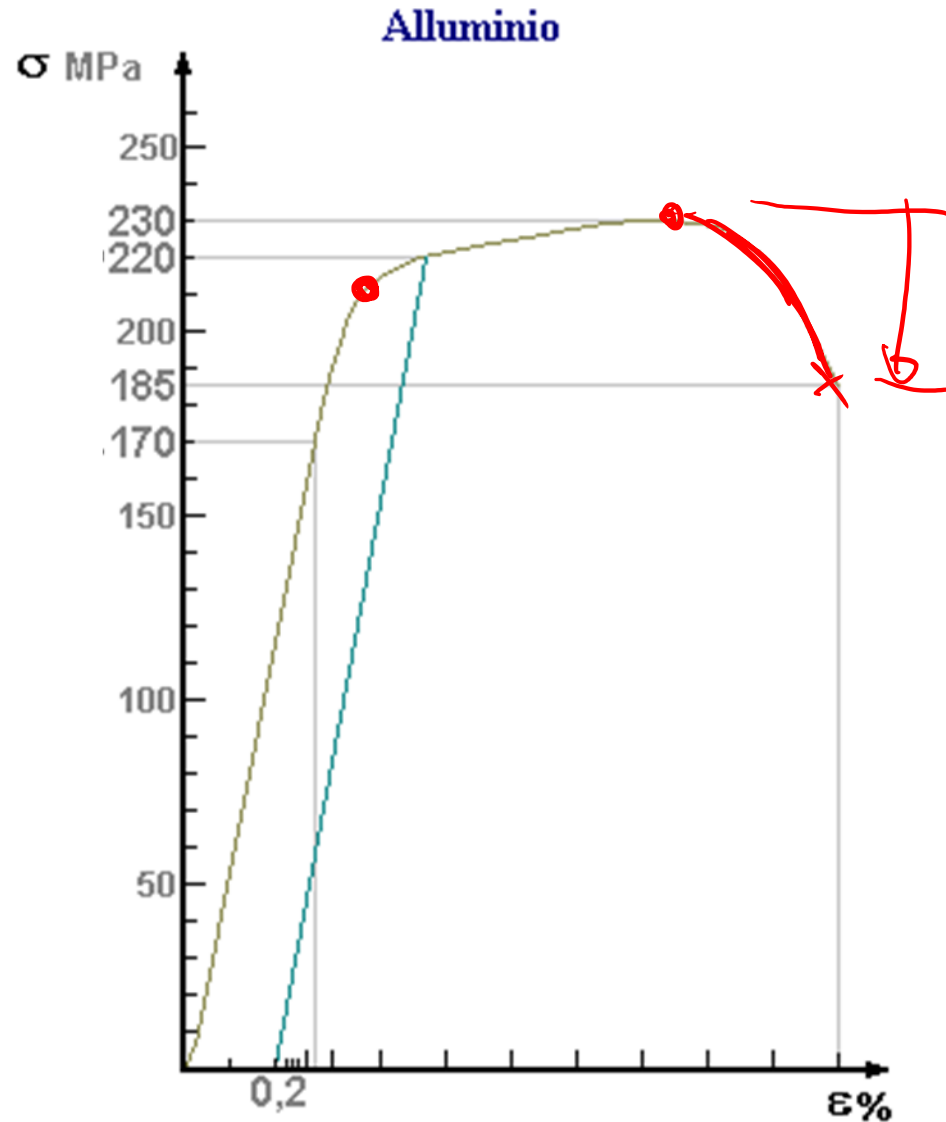
Quiz 3



The σ - ϵ curve in figure comes from the tensile test of an aluminum specimen. Select the right sentence (only one):

- a) The specimen had only elastic deformation
- b** ~~a~~) The specimen underwent necking
- c** ~~a~~) The yield strength was not reached in the test

Quiz 3



The σ - ϵ curve in figure comes from the tensile test of an aluminum specimen. Select the right sentence (only one):

- a) The specimen had only elastic deformation
- a) **The specimen underwent necking** ✓
- a) The yield strength was not reached in the test